FIVE PROPERTIES OF THE DIAGONAL MORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof recalls a computation performed in the proof of a cross-chapter lemma and then carries out the transfer for the relative diagonal explicitly, property by property.

Editorial note. The six references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The diagonal of an algebraic space over its base is representable with properties (2)-(5). Spaces, Lemma spaces-lemma-properties-diagonal, used at proof lines 397-400 and 432-434. Supplies the base case that the whole argument transfers.
- Properties of morphisms of algebraic spaces which are etale local on the source-and-target. Spaces, Lemma spaces-lemma-representable-morphisms-spaces-property, used at proof lines 429-432. Reduces the five claims to the corresponding claims for R -> U x_Y U.
- A composition of monomorphisms is a monomorphism and a monomorphism is separated. Schemes, Lemma schemes-lemma-monomorphism-separated, used at proof lines 440-442 and 449-450. Gives properties (3) and (4).
- Permanence of local finite typeness under composition. Morphisms, Lemma morphisms-lemma-permanence-finite-type, used at proof lines 443-445. Gives property (2) for R -> U x_Y U.
- A monomorphism locally of finite type has finite fibres, hence is locally quasi-finite. Morphisms, Lemma morphisms-lemma-finite-fibre, used at proof lines 446-448. Gives property (5).

Source reference. Original lemma 02NG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3479–3479. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01T8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2365–2365. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01L4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2073–2073. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02X4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3827–3827. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3822–3822. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\Delta_{X/Y} : X \to X \times_Y X$ be the diagonal morphism. Then*

(1) *$\Delta_{X/Y}$ is representable,*
(2) *$\Delta_{X/Y}$ is locally of finite type,*
(3) *$\Delta_{X/Y}$ is a monomorphism,*
(4) *$\Delta_{X/Y}$ is separated, and*
(5) *$\Delta_{X/Y}$ is locally quasi-finite.*

Proof. We are going to use the fact that $\Delta_{X/S}$ is representable (by definition of an algebraic space) and that it satisfies properties (2) – (5), see Spaces, Lemma 02X4. Note that we have a factorization

$$X \longrightarrow X \times_Y X \longrightarrow X \times_S X$$

of the diagonal $\Delta_{X/S} : X \to X \times_S X$. Since $X \times_Y X \to X \times_S X$ is a monomorphism, and since $\Delta_{X/S}$ is representable, it follows formally that $\Delta_{X/Y}$ is representable. In particular, the rest of the statements now make sense, see Section 03HA.

Choose a surjective étale morphism $U \to X$, with U a scheme. Consider the diagram

$$\begin{array}{ccccc} R = U \times_X U & \longrightarrow & U \times_Y U & \longrightarrow & U \times_S U \\ \downarrow & & \downarrow & & \downarrow \\ X & \longrightarrow & X \times_Y X & \longrightarrow & X \times_S X \end{array}$$

Both squares are cartesian, hence so is the outer rectangle. The top row consists of schemes, and the vertical arrows are surjective étale morphisms. By Spaces, Lemma 02WZ the properties (2) – (5) for $\Delta_{X/Y}$ are equivalent to those of $R \to U \times_Y U$. In the proof of Spaces, Lemma 02X4 we have seen that $R \to U \times_S U$ has properties (2) – (5). The morphism $U \times_Y U \to U \times_S U$ is a monomorphism of schemes. These facts imply that $R \to U \times_Y U$ have properties (2) – (5).

Namely: For (3), note that $R \to U \times_Y U$ is a monomorphism as the composition $R \to U \times_S U$ is a monomorphism. For (2), note that $R \to U \times_Y U$ is locally of finite type, as the composition $R \to U \times_S U$ is locally of finite type (Morphisms, Lemma 01T8). A monomorphism which is locally of finite type is locally quasi-finite because

it has finite fibres (Morphisms, Lemma 02NG), hence (5). A monomorphism is separated (Schemes, Lemma 01L4), hence (4). □